

Knowledge Control for Responsible Generative AI: Bridging Academia, Industry, and Society

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1 Tutorial Goal

Controlling the knowledge and behavior of generative AI systems, including large language models (LLMs), multimodal LLMs (MLLMs), and text-to-image (T2I) models, has become critical as they are increasingly used in safety-sensitive and socially impactful applications. These models often encode unintended, biased, or private content, leading to harmful or unethical outputs. Post-training knowledge control has thus emerged as a practical framework for selectively modifying or removing model behaviors without full retraining, offering scalable and interpretable interventions for improving safety, privacy, and fairness.

This tutorial introduces the foundations of post-training knowledge control and showcases recent frontier methods, bridging research insights with real-world practices from both academia and industry. We cover: (i) key motivations and failure modes, such as harmful generation and stereotype reinforcement; (ii) core methods such as machine unlearning, knowledge editing, and inference-time interventions for targeted behavior adjustment; and (iii) evaluation protocols for balancing forgetting, retention, and fairness. Case studies will span text and vision–language generation, including privacy preservation, bias mitigation, and factual correction. The topic is timely for the computational linguistics (CL) and natural language processing (NLP) community, as LLMs and MLLMs are increasingly used in multilingual, high-stakes settings where retraining is impractical. Attendees will gain a clear understanding of this emerging area and its practical methods for building more responsible and trustworthy generative models. This tutorial is classified as **cutting-edge in CL/NLP**.

2 Intended Audience

This tutorial is designed for researchers and practitioners in computational linguistics (CL), natural language processing (NLP), and responsible AI, as well as related areas such as machine learning and multimodal intelligence. While having prior experience with LLMs and transformer architectures will be helpful, it is not required for attending the tutorial. We will comprehensively cover the necessary background to make the material accessible to newcomers, while providing in-depth discussions and technical insights for more experienced participants. We anticipate an audience of approximately 50 to 100 attendees, including both academic researchers and industry professionals working on safe and deployable AI systems. After the tutorial, attendees will be able to (1) understand the conceptual foundations of post-training knowledge control, (2) apply methods such as machine unlearning, knowledge editing, and inference-time interventions to address issues of safety, fairness, and privacy, and (3) adapt these techniques to real-world NLP and multimodal applications, including dialogue systems, multilingual generation, and responsible deployment in practice.

3 Tutorial Outline

- **Introduction (20 min)**
 - Backgrounds and preliminaries
 - Motivations and taxonomy of post-training interventions
 - Academic and industry perspectives on responsible generative AI
- **Part I: Machine Unlearning for Knowledge Removal (30 min)**
 - Strategies for selective forgetting and behav-

ior suppression

- Balancing stability, retention, and scalability
- Practical implications for real-world deployment

- **Part II: Knowledge Editing for Targeted Modification (30 min)**

- Mechanistic and memory-based editing methods
- Factual correction, personalization, and reasoning consistency
- Extensions to multilingual and multimodal systems

- **Part III: Reflective and Bias-Aware Knowledge Control (30 min)**

- Bias detection and fairness-preserving interventions
- Reflection-based and self-alignment strategies
- Integrating fairness, faithfulness, and interpretability

- **Part IV: Evaluation and Responsible Deployment (30 min)**

- Metrics for forgetting–retention and fairness–utility balance
- Benchmarks for safety, privacy, and factual reliability
- Responsible deployment, auditing, and governance

- **Part V: Future Directions and Cross-Sector Discussion (40 min)**

- Scalable and certifiable unlearning for foundation models
- Bridging academia, industry, and society for trustworthy AI

4 Description of Topics

Introduction: We begin by introducing the concept of knowledge control in generative models. We re-define “knowledge” in the context of large-scale generative systems and distinguish knowledge control from related areas like prompt-based alignment or system-level guardrails. This topic is increasingly timely and essential as generative AI models, such as LLMs, MLLMs, and text-to-image (T2I) systems, are deployed in high-stakes applications where trustworthiness is critical. Recent incidents of harmful, biased, or privacy-violating outputs have underscored the urgency of scalable

and practical interventions. These real-world behaviors serve to ground the importance of model-level interventions and highlight the limitations of purely training-time or rule-based approaches. As such, post-training knowledge control has emerged as a promising framework for modifying or suppressing undesirable behaviors in deployed models without requiring full retraining.

Part I: Machine Unlearning for Knowledge Removal: This section will delve into algorithmic strategies aimed at removing unwanted knowledge from generative models post hoc. We will discuss techniques such as task vector subtraction (Ilharco et al., 2022; Liu et al., 2024), where directional updates in parameter space are used to subtract specific behaviors; gradient-ascent-based optimization, which introduces anti-training signals to reverse prior learning (Yao et al., 2024; Zhang et al., 2024); and in-context unlearning (Pawelczyk et al., 2023), a lightweight approach that uses tailored prompts to reduce the influence of specific examples at inference time. We will illustrate each method with examples across LLMs, MLLMs, and T2I models, and compare their trade-offs in terms of efficacy, retention of unrelated capabilities, and scalability to large foundation models.

Part II: Knowledge Editing for Targeted Modification: In this part, we turn to methods that modify rather than erase model knowledge. We will introduce knowledge editing techniques such as MEMIT (Meng et al., 2022b), memory patches (Murty et al., 2022) and ROME (Meng et al., 2022a). We will discuss their applicability across different model types, their ability to generalize to unseen prompts, and their role in real-world use cases like factual correction, personalization, and bias mitigation.

Part III: Knowledge Reflection and Distillation for Bias Mitigation: We will also cover other control strategies to resolve fairness-related problems in generative models. We will discuss bias-aware decoding (He et al., 2021), prompt-based intervention (Bansal et al., 2022), LLM feedback loop (Wan and Chang, 2024b,a), and distillation-based behavior modification (Dhar et al., 2021). For each approach, we will explain the intuition, method design, applicable model types, and key trade-offs in terms of scalability, stability and modalities, as well as their effectiveness in retaining general capabilities (e.g. reasoning) of models.

Part IV: Evaluation and Applications: Following this, we will examine how to evaluate the effectiveness and reliability of knowledge control methods. We introduce metrics for unlearning (e.g., divergence from original outputs), retention (e.g., fidelity on unrelated tasks), and trustworthiness-specific criteria such as fairness, safety, and privacy. We will also provide an overview of existing benchmarks, including datasets for stereotype evaluation (e.g. WinoBias (Zhao et al., 2018), Stereoset (Nadeem et al., 2021), Vlstereonet (Zhou et al., 2022), UniversalPersona (Wan et al., 2023), LABE (Wan and Chang, 2024b)), content safety assessment (e.g. WMDP (Li et al., 2024)) and privacy-sensitive generation tasks (e.g. TOFU (Maini et al., 2024), MLLMU-Bench (Liu et al., 2025), LUME (Ramakrishna et al., 2025)).

Part V: Future Directions and Open Challenges: This concluding part explores emerging challenges and cross-sector opportunities in knowledge control. We will examine the development of certifiable and scalable unlearning techniques, the limitations of current fairness-aware interventions, and the integration of post-training control into broader alignment and safety frameworks. Emphasizing collaboration across academia, industry, and society, the session connects theoretical advances with practical deployment and ethical governance. Industry experts will share perspectives on scaling unlearning and editing to frontier models, defining responsible boundaries for knowledge retention and removal, and establishing transparent, auditable standards for real-world systems. By synthesizing insights from researchers and practitioners, this discussion will highlight open problems, deployment challenges, and societal trade-offs, outlining a roadmap toward safer and more accountable generative AI.

5 Breadth of Inclusion

This tutorial incorporates a broad body of research on post-training knowledge control, integrating both the presenters’ own contributions and extensive work by other research groups across the NLP and ML communities. We will devote a fair amount of time to discussing others’ work, emphasizing how these studies lay the foundational principles, introduce key innovations, or inspire subsequent extensions within each topic area. Rather than presenting isolated results, the tutorial will trace how these contributions collectively shape the field’s un-

derstanding of controllable and trustworthy generative models. For each theme, ranging from machine unlearning and knowledge editing to bias mitigation and evaluation frameworks, we will explain how prior approaches complement one another, illustrating the evolution of ideas from conceptual formulations to practical algorithms. The presentation will highlight the interplay among works from academia and industry, showing how different lines of research converge toward the shared goal of ensuring safer and more interpretable language and multimodal models.

6 Reading List

The following readings provide foundational and representative coverage of post-training knowledge control methods. Papers marked with an asterisk (*) are recommended as pre-readings to help participants follow the technical material more easily.

• Part I: Machine Unlearning for Knowledge Removal

- *Yao, Yuanshun, Xiaojun Xu, and Yang Liu. “Large Language Model Unlearning.” NeurIPS, 2024.
- Lucki, Jakub, Boyi Wei, Yangsibo Huang, Peter Henderson, and Florian Tramèr. “An Adversarial Perspective on Machine Unlearning for AI Safety.” arXiv, 2024.

• Part II: Knowledge Editing for Targeted Modification

- *Meng, Kevin, David Bau, Alex Andonian, and Yonatan Belinkov. “Locating and Editing Factual Associations in GPT.” NeurIPS, 2022.
- Mitchell, Eric, Charles Lin, Antoine Bosselut, Christopher Manning, and Chelsea Finn. “Memory-Based Model Editing at Scale.” ICML, 2022.

• Part III: Knowledge Reflection and Distillation for Bias Mitigation

- *He, Zexue, Bodhisattwa Prasad Majumder, and Julian McAuley. “Detect and Perturb: Neutral Rewriting of Biased and Sensitive Text.” EMNLP Findings, 2021.
- Bansal, Hritik, Da Yin, Masoud Monajatipoor, and Kai-Wei Chang. “How Well

Can Text-to-Image Models Understand Ethical Natural Language Interventions?” EMNLP, 2022.

- **Part IV: Evaluation and Benchmarking in Safety, Fairness, and Privacy**

- *Maini, Pratyush, Zhili Feng, Avi Schwarzschild, Zachary C. Lipton, and J. Zico Kolter. “TOFU: A Task of Fictitious Unlearning for LLMs.” arXiv, 2024.
- Li, Nathaniel, Alexander Pan, Anjali Gopal, and others. “The WMDP Benchmark: Measuring and Reducing Malicious Use with Unlearning.” arXiv, 2024.

7 Prior Events

A related event, the *TrustNLP 2025* workshop (Cao et al., 2025), co-organized by two of our tutorial presenters (Yixin Wan and Prof. Kai-Wei Chang), attracted an audience of a similar size and demonstrated substantial community interest in responsible and trustworthy NLP. Building on that momentum, our tutorial extends the discussion from conceptual and empirical studies of trustworthiness to a systematic, hands-on treatment of post-training knowledge control in generative models.

8 Technical Requirements and Pedagogical Materials

The tutorial requires only standard audiovisual setup (projector, microphones, and internet) for presenting slides and live demonstrations. The tutorial will be delivered through prepared slides and pre-run examples, with interactive code available on Google Colab, requiring no additional computational resources.

To ensure long-term accessibility and educational reusability, all tutorial materials will be publicly available. These include slides, a GitHub repository with benchmark datasets and reference code, and Colab notebooks for interactive exploration. We will also provide classroom-ready materials, such as annotated slides and hands-on Colab exercises demonstrating key algorithms (e.g., unlearning and editing), suitable for use in university courses or workshops on trustworthy NLP and generative AI. An official tutorial website will host all resources, with video recordings and narrated slide decks provided for accessibility. None of the materials contain restricted content and can be released openly through the ACL Anthology.

9 Ethic Statement

The tutorial aims to address ethical considerations central to the development and deployment of large generative models, particularly those related to safety, fairness, privacy, and accountability. By focusing on post-training knowledge control, such as machine unlearning, knowledge editing, and bias mitigation, the tutorial directly contributes to responsible AI practices that aim to reduce harm, remove sensitive or biased knowledge, and enhance model transparency. All discussed methods will be presented with attention to their potential risks, including misuse of unlearning for censorship, over-removal that degrades model integrity, or biased evaluation metrics. We will emphasize the importance of ethical experimentation, responsible dataset handling, and transparent reporting of interventions. Overall, the tutorial seeks to promote technically rigorous and ethically sound approaches for ensuring trustworthy language and multimodal AI systems.

10 Diversity Statement

Our tutorial team reflects diversity in gender, career stage, institution, and research background, with members from both academia and industry, including the University of Notre Dame, UCLA, AI2, Anthropic, Meta AI, and USC. The team spans Ph.D. students, early-career faculty, senior scientists, and industry experts in safety, fairness, and alignment, bringing together emerging and established perspectives toward responsible generative AI. This mix ensures balanced viewpoints across research and deployment contexts. The tutorial also promotes linguistic and domain diversity by showing how post-training control methods extend to multilingual and multimodal settings. All slides, code, and recordings will be publicly released and shared through diversity-oriented NLP communities (e.g., Women in NLP, Black in AI, Masakhane), fostering inclusive participation and collaboration across sectors.

11 Venue Compatibility

This tutorial suits both ACL and EACL, addressing broad methodological and ethical challenges relevant to the global CL/NLP community.

Presenter Biography

Zheyuan (Frank) Liu is a Ph.D. student in Computer Science and Engineering at the University of Notre Dame. He received dual B.S. degrees in Computer Science and Applied Mathematics from Brandeis University. His research focuses on post-training methods for trustworthy generative AI, including machine unlearning, safety alignment, and fairness in LLMs and MLLMs. One of his unlearning works has been granted a U.S. patent. He has presented at major international conferences including [ACL](#), [NAACL](#), and [WWW](#).

Yixin Wan is a Ph.D. student in Computer Science at the University of California, Los Angeles. She obtained dual B.S. degrees in Applied Mathematics and Economics from UCLA. Her research focuses on evaluating and improving the trustworthiness of multimodal generative models, revealing new risks related to fairness, faithfulness, and reasoning in large-scale AI systems via techniques such as unlearning and alignment. One of her papers received the Best Short Paper Award at the TrustNLP workshop at [NAACL 2024](#). She has presented her work at international venues including [EMNLP](#), [ACL](#), and [NAACL](#).

Kai-Wei Chang is an Associate Professor of Computer Science at the University of California, Los Angeles. Before joining UCLA, he was a faculty member at the University of Virginia and a Postdoctoral Researcher at Microsoft Research New England. He received his Ph.D. in Computer Science from the UIUC in 2015. His research focuses on trustworthy NLP, multimodal foundation models, and reasoning in NLP. He has received numerous honors, including the Sloan Research Fellowship, [AAAI](#) Senior Member recognition, and multiple Best Paper Awards at [EMNLP](#), [KDD](#), and [ACL](#). He has delivered tutorials at major international venues such as [EMNLP](#), [AAAI](#) and [ACL](#), and has organized the Trustworthy NLP workshop series across top conferences.

Meng Jiang is Frank M. Freimann Collegiate Professor of Computer Science & Engineering at the University of Notre Dame. He holds a B.E. and Ph.D. from Tsinghua University and completed his post-doctoral research at the UIUC. His research spans data mining, machine learning, and NLP, with recent interests in AI ethics, personalized LLMs, and machine unlearning. He has delivered

multiple tutorials and invited talks at international venues including [EMNLP](#), [ACL](#), and [WSDM](#).

Nouha Dziri is a Research Scientist at the Allen Institute for AI (AI2), where she works on safety and post-training techniques for [OLMo](#), an open LLM advancing responsible AI. Her research explores the limitations and reasoning capabilities of Transformer-based models, with focus on alignment, interpretability, and reward modeling. Her recent contributions include [Faith and Fate](#), [RewardBench](#), and [Finegrained RLHF](#). She has presented tutorials and workshops at major international venues including [ICML](#), [NeurIPS](#), and [ICLR](#).

Jieyu Zhao is an Assistant Professor in the Department of Computer Science at the University of Southern California. Prior to this, she was an NSF Computing Innovation Fellow at the University of Maryland, College Park. She received her Ph.D. in Computer Science from UCLA. Her research focuses on fairness in machine learning and NLP. She was selected to participate in the 2021 Rising Stars in [EECS](#) workshop. Her research has garnered attention from major media outlets, including [Wired](#) and [The Daily Mail](#). She has delivered talks to a wide range of international audiences at various venues, including [AAAI](#), [NAACL](#), and [IJCAI](#).

Yuning Mao is a Research Scientist at Meta SuperIntelligence Labs and a founding member of the Meta GenAI organization. As part of the Llama Post-training team, he has contributed to Llama 2, 3, and 4. His recent work focuses on code generation with LLMs, building on prior contributions to LLM safety, including safety evaluations for Llama 2, multilingual and internationalization safety for Llama 3, and Meta AI's automated red-teaming framework. He received his Ph.D. in Computer Science from UIUC and has delivered talks at major venues such as [EMNLP](#), [ACL](#), and [KDD](#).

References

- Hritik Bansal, Da Yin, Masoud Monajatipoor, and Kai-Wei Chang. 2022. How well can text-to-image generative models understand ethical natural language interventions? In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 1358–1370.
- Trista Cao, Anubrata Das, Tharindu Kumarage, Yixin Wan, Satyapriya Krishna, Ninareh Mehrabi, Jwala Dhamala, Anil Ramakrishna, Aram Galystan, Anoop Kumar, Rahul Gupta, and Kai-Wei Chang, editors.

2025. *Proceedings of the 5th Workshop on Trustworthy NLP (TrustNLP 2025)*. Association for Computational Linguistics, Albuquerque, New Mexico.
- Prithviraj Dhar, Joshua Gleason, Aniket Roy, Carlos D Castillo, P Jonathon Phillips, and Rama Chellappa. 2021. Distill and de-bias: Mitigating bias in face verification using knowledge distillation. *arXiv preprint arXiv:2112.09786*.
- Zexue He, Bodhisattwa Prasad Majumder, and Julian McAuley. 2021. Detect and perturb: Neutral rewriting of biased and sensitive text via gradient-based decoding. *arXiv preprint arXiv:2109.11708*.
- Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hananeh Hajishirzi, and Ali Farhadi. 2022. Editing models with task arithmetic. *arXiv preprint arXiv:2212.04089*.
- Nathaniel Li, Alexander Pan, Anjali Gopal, Summer Yue, Daniel Berrios, Alice Gatti, Justin D Li, Ann-Kathrin Dombrowski, Shashwat Goel, Long Phan, and 1 others. 2024. The wmdp benchmark: Measuring and reducing malicious use with unlearning. *arXiv preprint arXiv:2403.03218*.
- Zheyuan Liu, Guangyao Dou, Zhaoxuan Tan, Yijun Tian, and Meng Jiang. 2024. Towards safer large language models through machine unlearning. *ACL Findings*.
- Zheyuan Liu, Guangyao Dou, Xiangchi Yuan, Chunhui Zhang, Zhaoxuan Tan, and Meng Jiang. 2025. Modality-aware neuron pruning for unlearning in multimodal large language models. *ACL Main*.
- Pratyush Maini, Zhili Feng, Avi Schwarzschild, Zachary C Lipton, and J Zico Kolter. 2024. Tofu: A task of fictitious unlearning for llms. *arXiv preprint arXiv:2401.06121*.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2022a. Locating and editing factual associations in gpt. *Advances in neural information processing systems*, 35:17359–17372.
- Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. 2022b. Mass-editing memory in a transformer. *arXiv preprint arXiv:2210.07229*.
- Shikhar Murty, Christopher D Manning, Scott Lundberg, and Marco Tulio Ribeiro. 2022. Fixing model bugs with natural language patches. *arXiv preprint arXiv:2211.03318*.
- Moin Nadeem, Anna Bethke, and Siva Reddy. 2021. *StereoSet: Measuring stereotypical bias in pretrained language models*. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 5356–5371, Online. Association for Computational Linguistics.
- Martin Pawelczyk, Seth Neel, and Himabindu Lakkaraju. 2023. In-context unlearning: Language models as few shot unlearners. *arXiv preprint arXiv:2310.07579*.
- Anil Ramakrishna, Yixin Wan, Xiaomeng Jin, Kai-Wei Chang, Zhiqi Bu, Bhanukiran Vinzamuri, Volkan Cevher, Mingyi Hong, and Rahul Gupta. 2025. Lume: Llm unlearning with multitask evaluations. *arXiv preprint arXiv:2502.15097*.
- Yixin Wan and Kai-Wei Chang. 2024a. The male ceo and the female assistant: Probing gender biases in text-to-image models through paired stereotype test. *arXiv e-prints*, pages arXiv–2402.
- Yixin Wan and Kai-Wei Chang. 2024b. White men lead, black women help: Uncovering gender, racial, and intersectional bias in language agency. *arXiv e-prints*, pages arXiv–2404.
- Yixin Wan, Jieyu Zhao, Aman Chadha, Nanyun Peng, and Kai-Wei Chang. 2023. Are personalized stochastic parrots more dangerous? evaluating persona biases in dialogue systems. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 9677–9705.
- Yuanshun Yao, Xiaojun Xu, and Yang Liu. 2024. Large language model unlearning. *Advances in Neural Information Processing Systems*, 37:105425–105475.
- Ruiqi Zhang, Licong Lin, Yu Bai, and Song Mei. 2024. Negative preference optimization: From catastrophic collapse to effective unlearning. *arXiv preprint arXiv:2404.05868*.
- Jieyu Zhao, Tianlu Wang, Mark Yatskar, Vicente Ordonez, and Kai-Wei Chang. 2018. Gender bias in coreference resolution: Evaluation and debiasing methods. *arXiv preprint arXiv:1804.06876*.
- Kankan Zhou, Yibin LAI, and Jing Jiang. 2022. V1-stereoset: A study of stereotypical bias in pre-trained vision-language models. Association for Computational Linguistics.